6.	Solid	Gas	Solid foam	Pumice stone, Foam rubber bread.
7.	Solid	Liquid	Gel	Butter, cheese
8.	Solid	Solid	Solid sol	Pearls, opals coloured glass alloys colloidal dispersed eutetics.

10.5.2 Preparation of Colloids

Many lyophillic substances are made in their colloidal form by warming with water. Rubber forms colloidal solution with benzene. Soap spontaneously forms a colloidal solution by just mixing with water.

In general, colloidal are prepared by the following methods.

i. Dispersion methods: In this method larger particles are broken to colloidal dimension.

ii. Condensation method: In this method, smaller atom or molecules are converted into larger colloidal sized particles.

1) Dispersion methods

(i) Mechanical Dispersion:

Using a colloid mill, the solid is ground to colloidal dimension. The colloid mill consists of two metal plates rotating in opposite direction at very high speed of nearly 7000 revolution / minute.

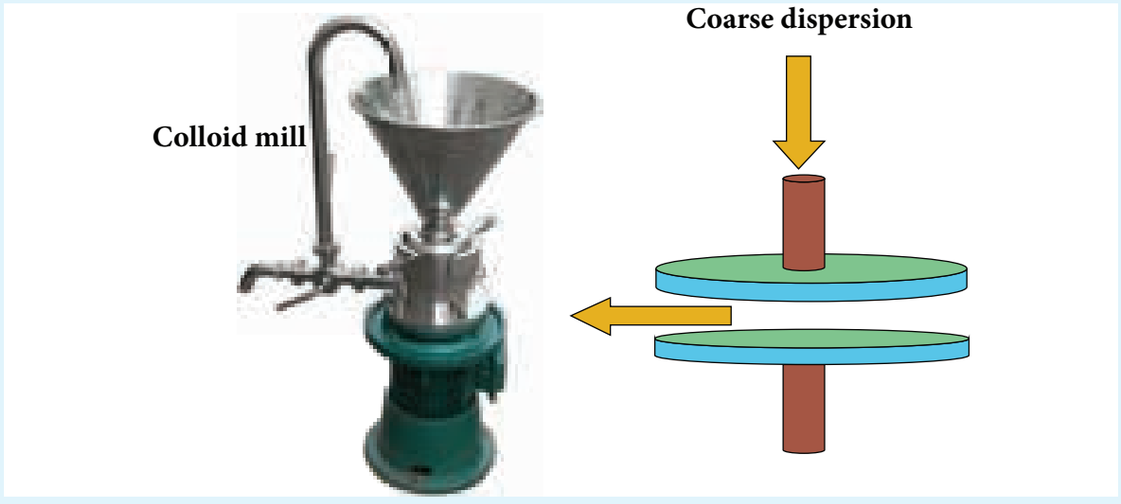


Figure 10.8 Colloid mill

The colloidal particles of required colloidal size is obtained by adjusting the distance between two plates.

By this method, colloidal solutions of ink and graphite are prepared.

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